

~~Equivalent DFA~~

Minimization of FSM using partition method.

The minimization of FSM means reducing the no. of states ~~from~~ⁱⁿ given FA.

Method for construction of Minimum State automata

Suppose there is a DFA $D = \langle Q, \Sigma, q_0, \delta, f \rangle$ which recognizes a lang L . Then the minimized DFA $D' = \langle Q', \Sigma, q_0', \delta', f \rangle$ can be constructed for Lang L as

Step 1: We will divide Q (Set of states) into two sets. One set will contain all final states & other set will contain non-final states. This partition is called P_0 .

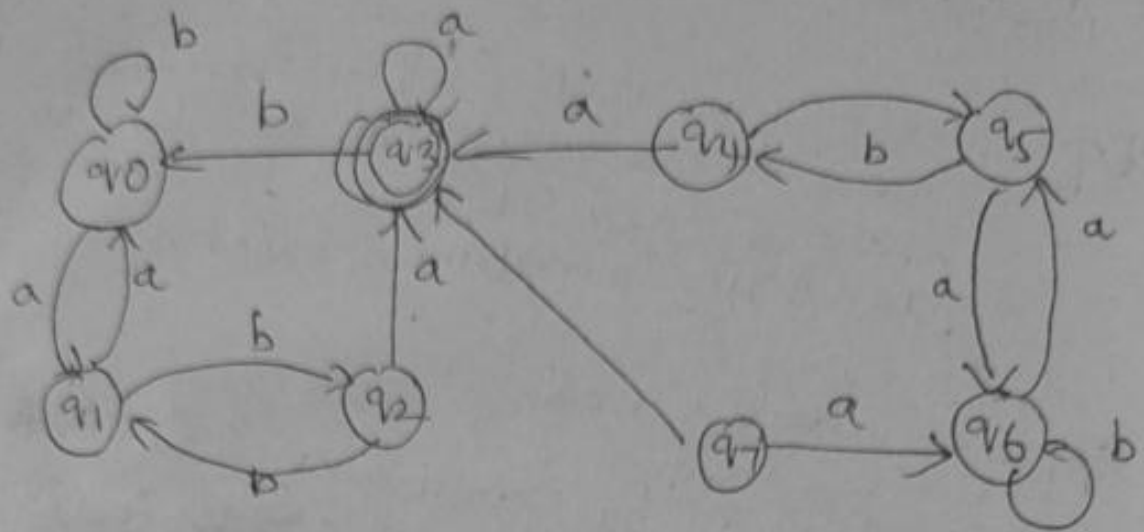
Step 2: initialize $k=1$

Step 3: find P_k by partitioning the different sets of P_{k-1} . In each set of P_{k-1} , we will take all possible pairs of states. If 2 states of a set are distinguishable we will split the sets into different sets in P_k .

Step 4: Stop when $P_k = P_{k-1}$ (No change in partition)

Step 5: All states of one set are merged into one. No. of states in minimized DFA will be equal to no. of sets in P_k .

Q) Construct the minimum state automata for the following transition diagram.



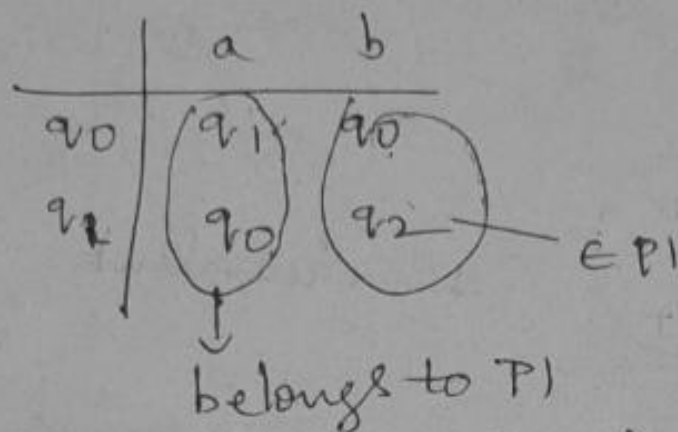
Transition table

	a	b
q ₀	q ₁	q ₀
q ₁	q ₀	q ₂
q ₂	q ₃	q ₁
q ₃	q ₃	q ₀
q ₄	q ₃	q ₅
q ₅	q ₆	q ₄
q ₆	q ₅	q ₆
q ₇	q ₆	q ₃

① $P_0 = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$
 $P_1 = \{q_0, q_1, q_2, q_4, q_5, q_6, q_7\}$
 $P_2 = \{q_3\}$

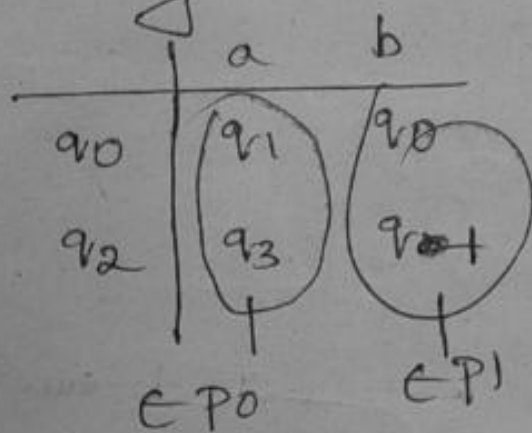
Partition P_1

Consider

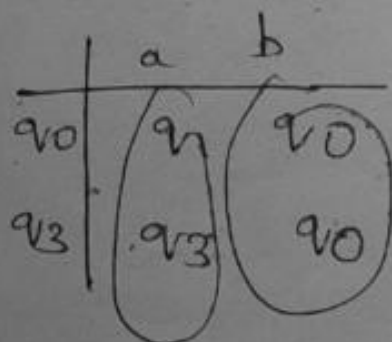


both are equivalent i.e. from the same set.

Hence they are 1 equivalent.



Hence they are not 1-equivalent



not 1-equivalent

	a	b
q_0	q_1	q_0
q_4	q_3	q_5

not 1-equivalent

	a	b
q_0	q_1	q_0
q_5	q_6	q_4

1-equivalent

	a	b
q_0	q_1	q_0
q_6	q_5	q_6

1-equivalent

	a	b
q_0	q_1	q_0
q_7	q_6	q_3

not 1-equivalent

1-equivalent
 $P_3 = \{q_0, q_1, q_5, q_6\}$

$P_2 = \{q_5\}$
 $P_4 = \{q_2, q_4, q_7\}$

$P_4 = \{q_2, q_4, q_7\}$

	a	b
q_2	q_3	q_1
q_4	q_5	q_5

1-equivalent

not 1-equivalent

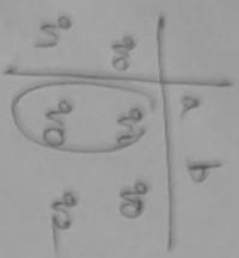
	a	b
q_2	q_3	q_1
q_7	q_6	q_3

$P_5 = \{q_2, q_4\}$

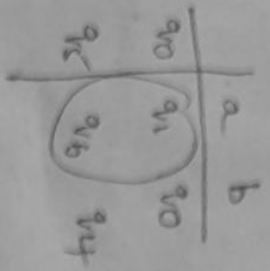
$P_6 = \{q_2\}$

Iteration₀ = $\{ [q_3] [q_0, q_1, q_5, q_6] [q_2, q_4] [q_7] \}$

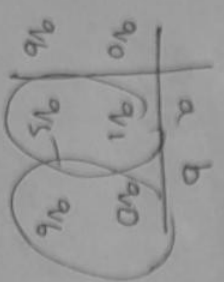
$\{q_0, q_1, q_5, q_6\}$



not ~~A~~ equivalent



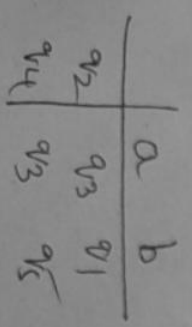
not ~~A~~ equivalent



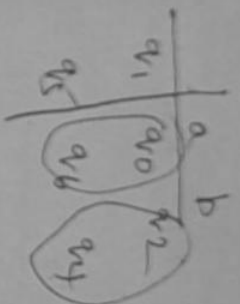
~~not~~ equivalent

$P_1 = \{q_0, q_6\}$ $P_2 = \{q_1, q_5\}$

$P_5 = \{q_2, q_4\}$



~~A~~-equivalent

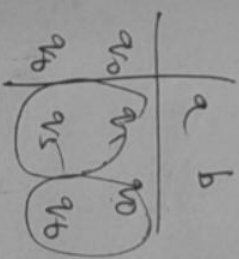


~~A~~-equivalent

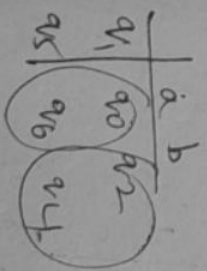
$$\text{Iteration 2} = [q_3][q_0, q_6][q_1, q_5][q_2, q_4][q_7]$$

Iteration 3

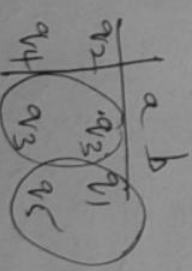
q₀ q₆



3-equivalent



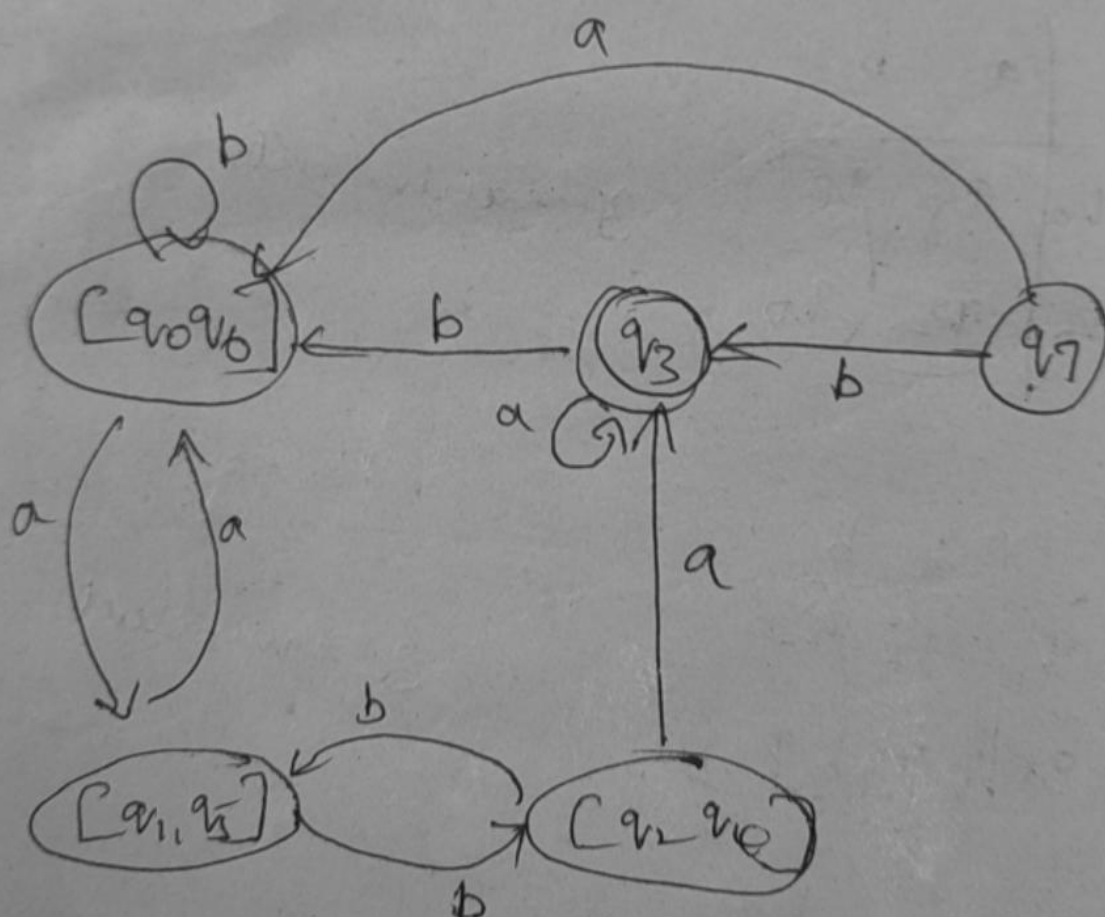
3-equivalent



3-equivalent

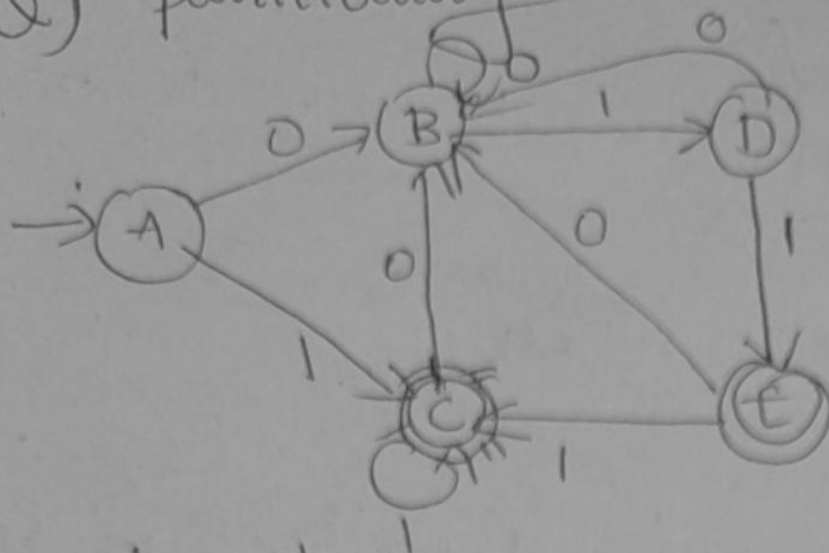
no more iterations

	a	b
$q_0 q_6$	$q_1 q_5$	$q_0 q_6$
$q_1 q_5$	$q_0 q_6$	$q_2 q_4$
$q_2 q_4$	q_3	$q_1 q_5$
q_3	q_3	$q_0 q_6$
q_7	$q_0 q_6$	q_3



$\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$

Q) partitioning method



	0	1
→ A	B	C
B	B	D
C	B	C
D	B	(E)
(E)	B	C

0-equivalence = {A, B, C, D} {E} non final
 1-equivalence = {A, B, C} {D, E}

	0	1
A	B	C
B	B	D

Same state
1-equivalence

same set

1-equivalent

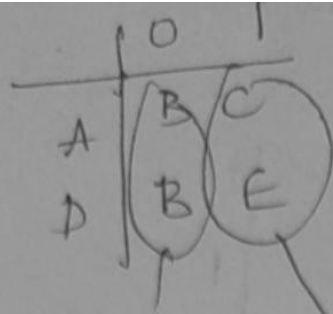
	0	1
A	B	C
C	B	C

same state

same state

A, C
equivalent

1-equivalent

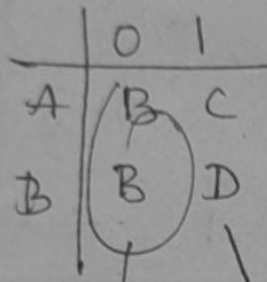


not 1-equivalent

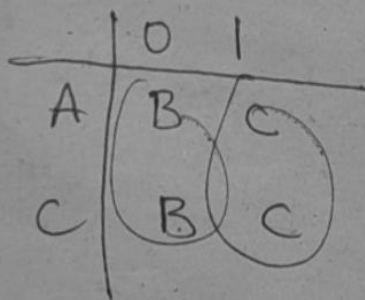
$$1\text{-equivalence} = \{A, B, C\} \{D\} \{E\}$$

Check for 2-equivalence = ?

$$\{A, C\} \{B\} \{D\} \{E\}$$

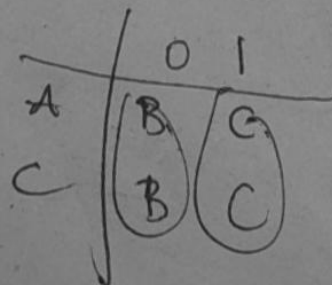


A, B are
not 2-equivalent

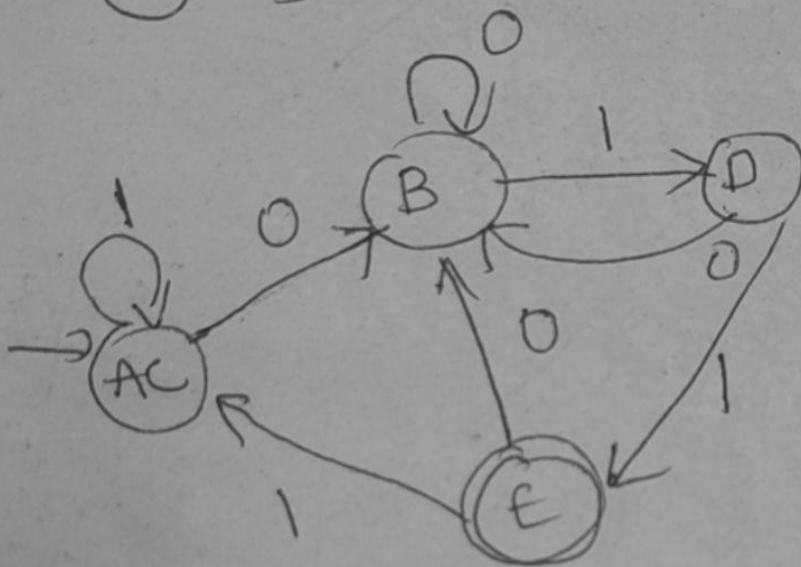


2-equivalent

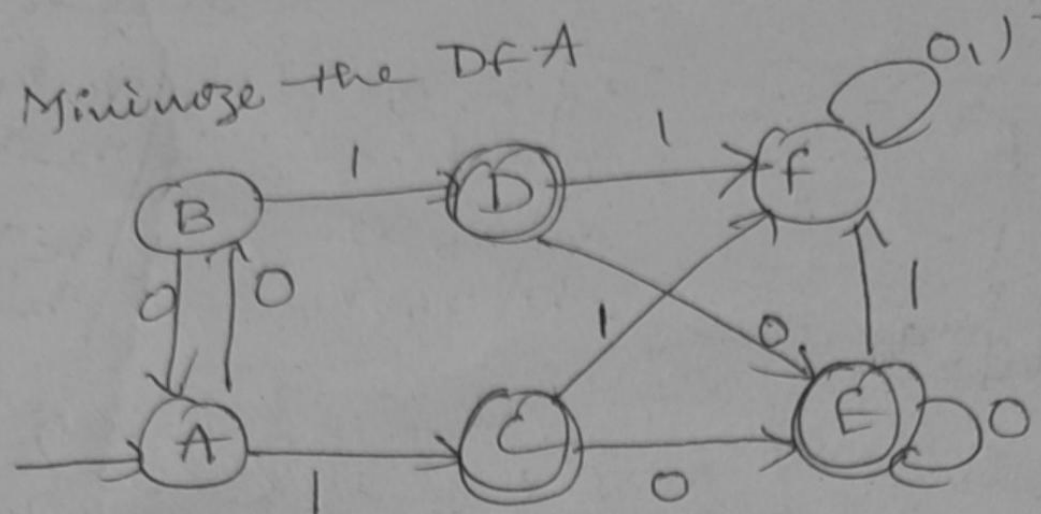
$$3\text{-equivalence} = \{A, C\} \{B\} \{D\} \{E\}$$



	0	1
→ A	B	C
B	B	D
C	B	C
D	B	E
⊕ E	B	C



0) Minimize the DFA



0-equivalence

$\{A, B, f\} \quad \{C, D, E\}$

1-equivalence

$\{A, B\} \quad \{f\} \quad \{C, D, E\}$

2-equivalence

$\{A, B\} \quad \{f\} \quad \{C, D, E\}$

	0	1
A	B	C
B	A	D
C	E	f
D	E	f
E	E	f
f	f	f

A	B	C	
B	A	D	✓
A	B	C	
f	f	f	X

$(AB) (f) (CDE)$

	0	1
AB	$\{AB\}$	$\{CDE\}$
f	$\{f\}$	$\{f\}$
CDE	$\{CDE\}$	$\{f\}$

